MSG No. 82.40 $I\bar{4}1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4..1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,7,8,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,7,8,9,10,15,16]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,4,5,6,11,12,13,14]`

Table 4: Wyckoff site: `2d`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[1,2,7,8,9,10,15,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,4,5,6,11,12,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `2..1'`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,9,10]`
2	$[0, 0, -z]$	`[3,4,11,12]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[5,6,13,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7,8,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,2,9,10]`
2	$\left[\frac{1}{2}, 0, -z\right]$	`[3,4,11,12]`
3	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[5,6,13,14]`
4	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7,8,15,16]`

Table 7: Wyckoff site: 8g, site symmetry: 11'

No.	position	mapping
1	$[x,\, y,\, z]$	[1,9]
2	$[-x,\, -y,\, z]$	[2,10]
3	$[y,\, -x,\, -z]$	[3,11]
4	$[-y,\, x,\, -z]$	[4,12]
5	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[5,13]
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[6,14]
7	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[7,15]
8	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[8,16]